

Corporate Bankruptcy Prediction using Machine Learning

UE24CS352A - Machine Learning | Mini-Project Write-up | PES University

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1. Problem Statement

Predict whether a company will go bankrupt from its financial ratios. This is a binary classification task (Bankrupt = 1, Non-Bankrupt = 0). Bankrupt firms are rare, so a model that always predicts "healthy" is already about 95% accurate yet useless. The goal is to build and tune several classifiers, compare them with imbalance-aware metrics (PR-AUC, recall, F1), and select the best model without using the test set for selection.

2. Dataset

UCI Polish Companies Bankruptcy dataset (id 365), fetched at run time with the *ucimlrepo* package, so no manual download is needed. Each record has 64 financial ratios (A1 to A64, e.g. A1 = net profit / total assets, A27 = operating profit / financial expenses) plus a *year* column identifying the forecasting horizon (1 to 5). Bankrupt firms were observed in 2000-2012 and operating firms in 2007-2013.

After removing exact duplicate rows, **43,004 records** remain: 2,083 bankrupt (4.84%) and 40,921 non-bankrupt (about 19.6 : 1). Data issues found: missing values (A37 about 44%, A21 about 14%, A27 about 6%), infinite ratios from division by zero, heavy-tailed outliers and strongly correlated profitability ratios. A stratified 80/20 split gives 34,403 training rows and 8,601 test rows (417 bankrupt).

3. Approach

Pipeline: load, inspect, drop duplicates, EDA, stratified split, per-model scikit-learn Pipeline with RandomizedSearchCV, test evaluation, comparison, interpretation, save best model.

- **Preprocessing** (inside each pipeline, refit per CV fold, so no leakage): infinity to NaN, median imputation, removal of constant columns; RobustScaler and clipping to ± 10 for the linear models only. Outliers are analysed, not removed.
- **Imbalance:** class weights ("balanced") instead of undersampling, oversampling or SMOTE, so no data is removed or synthesised.
- **Models:** Logistic Regression, Decision Tree, Random Forest, Linear SVM and Gradient Boosting (HistGradientBoosting).
- **Tuning and selection:** RandomizedSearchCV with stratified 5-fold CV on training data only (up to 6 candidates per model), scored by PR-AUC. The final model is chosen by cross-validated PR-AUC; the test set is used once, for the final report.

4. Implementation Overview

Python 3 with pandas, NumPy, scikit-learn, matplotlib and joblib. The code is organised as modules in *src/*:

- *data_loader.py*: fetch, validate, inspect and de-duplicate the data.
- *eda.py*: class balance, missing values, Spearman correlation heatmap, feature distributions.
- *preprocessing.py*: preprocessing pipeline and imbalance analysis.
- *train.py*: model grids, train/test split, tuning and saving the best model.
- *evaluate.py*: metrics, confusion matrices, ROC/PR curves, comparison table and interpretation.
- *predict.py*: command-line prediction from CSV or JSON, plus a template generator.

A Jupyter notebook reproduces the analysis. Running *python src/train.py* saves metrics and figures to *results/* and the best pipeline to *models/best_model.joblib* with metadata; setup steps are in the README.

5. Results and Conclusions

Test-set results (8,601 firms; precision, recall and F1 at the default 0.5 threshold):

Model	Precision	Recall	F1	ROC-AUC	PR-AUC (test)	PR-AUC (CV)
Gradient Boosting	0.707	0.712	0.710	0.968	0.800	0.836
Random Forest	0.783	0.345	0.479	0.927	0.583	0.626
Decision Tree	0.233	0.681	0.347	0.827	0.544	0.537
Logistic Regression	0.114	0.626	0.192	0.761	0.151	0.170
Linear SVM	0.115	0.621	0.194	0.758	0.154	0.170

- **Gradient Boosting is the best model:** highest PR-AUC (0.800) and ROC-AUC (0.968), with precision, recall and F1 all about 0.71. It caught 297 of 417 bankrupt firms with 123 false alarms among 8,184 healthy firms.
- Random Forest has the highest precision but misses most bankruptcies (recall 0.345). Logistic Regression and Linear SVM catch about 62% of bankruptcies but flag about 25% of healthy firms, likely because straight-line boundaries cannot capture non-linear ratio effects.
- Cross-validated and test PR-AUC are close for every model, so the selection is stable. Accuracy alone would have been misleading.
- Permutation importance shows the most influential features are A27 (operating profit / financial expenses), A46 (quick ratio), A34 (operating expenses / total liabilities), A58 (total costs / total sales) and A5, i.e. debt-cost coverage, liquidity and cost efficiency.
- **Limitations and future work:** fixed 0.5 threshold (threshold tuning could raise recall), a random row-level split, and data from one region and period. Next steps: threshold tuning, SHAP explanations, resampling methods and newer data.

6. Team Members and Contributions

Team member	Contributions
Nidhi Karkada PES2UG24CS314	Overall ML pipeline, model implementation, evaluation, final integration, live demo execution
Nidhi Raghuraman PES2UG24CS315	Dataset research, literature review, EDA, visualization, documentation support
Piyush A Patel PES2UG24CS350	Preprocessing validation, model comparison, GitHub organization, README, presentation preparation